

Treat the following exercises:

Exercise 1:

(3pts)

Given the following XML document "Collection.xml":

```
<?xml version="1.0" standalone="yes" ?>
```

1. <collection>
2. <housePlant>
3. <common> blood&root </common>
4. <botanical> sanguinary canadensis </botanical>
5. <zone num= 4/><zone>
6. <light> mostly shady</light>
7. <price currency="€" >2.44<5.00</price>
8. <availability> "599" </availability>
9. </houseplant>
10. </Collection>
11. <xml_plant>
12. <common>columbine</common>
13. <botanical>aquilegia's canadensis</botanical>
14. <zone num="3"/>
15. <light>mostly shady
16. <price currency="\$" >9.37>10.5
17. </light>
18. </price>
19. <availability>699</availability>
20. </xml_plant>
21. <houseplant>
22. <common>marsh marigold</common>
23. <botanical>caltha palustris
24. <zone num="4"/>
25. <light>mostly sunny</light>
26. <price currency="\$">6.81</price>
27. <availability>799</availability>
28. </houseplant>
29. </collection>

The XML document "Collection.xml" is not well formed:

- For each error, indicate the line number and explain the cause of this error.
- For each error propose a correction.

Exercise 2:

(5pts)

A. Write the schema file for the following XML document (drugs.xml).

B. In the XML document, add the necessary attributes to the root element which insures the link of the XML file to your schema file.

```
<?xml version="1.0" standalone="yes" ?>
<drugs>
  <drug>
    <name> Panadol </name>
    <content> Paracetamol 10 ml /150mg </content>
    <available> Yes </available>
    <analogon> adol</analogon>
  </drug>
  <vitamin>
    <name> 600 D </name>
    <content> Calcium V D3 2100 mg 200 IU </content>
    <available> Yes </available>
    <analogon> D3</analogon>
  </vitamin>
  <vitamin>
    <name> B12 </name>
    <content> V B12 150 IU </content>
    <available> Yes </available>
    <analogon> V 150 B12 +</analogon>
  </vitamin >
  <drug>
    <name> Prospan </name>
    <content> sugar sorbitol 2.5 ml /0.963 g 0.08 BU </content>
    <available> Yes </available>
    <analogon> Sorbitol</analogon>
  </drug>
</drugs>
```

Exercise 3:

(5pts)

Write the corresponding DTD file for the following XML document's description:

- The file name is Tennis_match.xml.
- For each match there is: two players, a date, a time and a playground
- The two players have different tag names, player1 and player2.
- Each player has the following information: a name, a birth_date, a nationality, a telephone_number and a previous_cup
- If a previous_cup exists, the year (attribute) and the country must be specified.

Exercise 4:

(4pts)

Write an instance XML document for the DTD file defined in exercise 3 (only for two matches).

Exercise 5:

(3pts)

Write a CSS file for the XML instance in exercise 4 to do the following formatting on the contents of tags:

- All font sizes are 14pt.
- The name of the player is bold.
- The playground is in double border.
- The color of the previous_cup is red.
- The birthdate is in italic.
- The match date is underlined.
- The background color is yellow.
- The font characters of the player names and playground is Times New Roman.